Geophys. J. Int. (1991) **107**, 333–351

Tectonic rotations in the Late Palaeozoic continental margin of southern South America determined and dated by palaeomagnetism

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Accepted 1991 May 30. Received 1991 May 22; in original form 1990 October 11

SUMMARY

Detailed palaeomagnetic studies of Late Palaeozoic rocks exposed in the Uspallata–Calingasta Valley region, located at about 32°S 69.5°W in the Argentine Andean Chain, have been recently accomplished. Previous results are discussed in Valencio & Vilas (1985) and Rapalini *et al.* (1989). The most recent studies were carried out mainly on the thick sequences of late Early Permian to Late Permian rhyolites and ignimbrites assigned to the Tambillos and Horcajo Formations. Both formations yielded palaeomagnetic pole positions (P15: 319.6°E, 78.9°S, $N = 16$, $k = 33.5$, $A_{95} = 6.5°$ and P17: 264.8°E, 72.4°S, $N = 26$, $k = 6.4$, $A_{95} = 12°$, respectively) concordant with the Late-Palaeozoic path of South America. Preliminary palaeomagnetic results from the Middle Carboniferous Hoyada Verde Formation, exposed in the same region, suggest that these rocks carry a synfolding magnetization acquired during the Late Carboniferous, as the palaeomagnetic pole position computed from partially corrected remanence directions (C6: 356.2°E, 41.9°S, $D1 = 8.3°$, $D2 = 6.0°$) agrees with the Late Carboniferous poles from cratonic areas of South America. This concordant position of C6 definitely rules out the possibility of a Late Palaeozoic allochthony of this section of the Argentine Andean Chain. The discordant positions previously found in Late Carboniferous and early Early Permian rocks of this region are interpreted as caused by large clockwise crustal block rotations that occurred not later than the late Early Permian. A tectonic model is proposed to explain these rotations which suggests that they were associated with strike–slip displacements parallel to the western continental margin of South America caused by an oblique subduction of the Proto-Pacific Plate during the Early Permian. Many tectonic and geologic observations fit this model. The concordant position of C6 also suggests that a complex pattern of crustal block rotations should have developed in this continental margin. Other palaeomagnetic data suggest that similar rotations may have taken place in other areas of the Argentine Andean Chain in the Early Permian.

Key words: Gondwana, Late Palaeozoic, oblique subduction, palaeomagnetism, Southern Andes, tectonic rotations.

1 INTRODUCTION

Recent palaeomagnetic studies on Late Palaeozoic rocks from the Southern Andes, in Argentina, have yielded palaeomagnetic pole (PP) positions that are not coincident with the apparent polar wandering path (APWP) of South America (Valencio & Vilas 1985; Rapalini *et al.* 1989). These 'anomalous' PP positions have been interpreted in terms of two different geodynamic hypotheses. One suggested *in situ* crustal block rotations while the other proposed the existence of a Late Palaeozoic displaced terrane in the Southern Andes. Most of these 'anomalous' PPs belong to Late Palaeozoic rocks exposed in the Uspallata–Calingasta Valley region, located at about 32°S, 69.5°W (Fig. 1). New palaeomagnetic studies were carried

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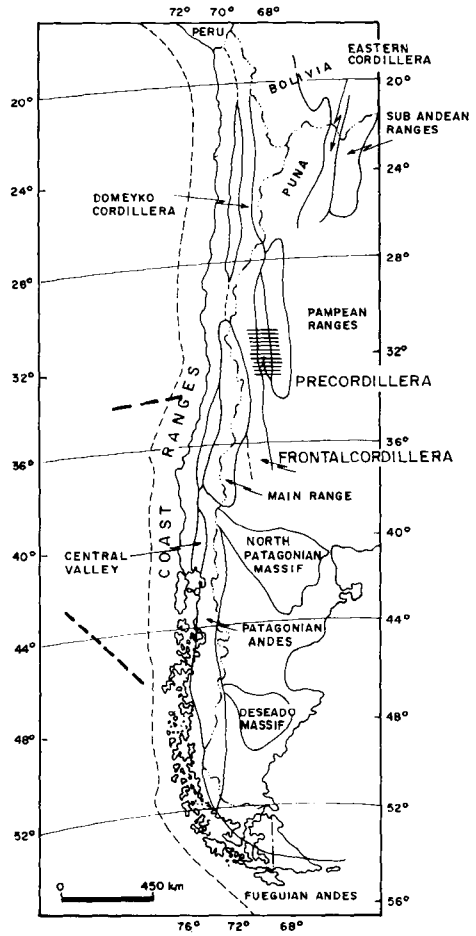


Figure 1. Main morphotectonic provinces of Southern South America and location of the Uspallata–Calingasta Valley region (shaded area). From Hervé *et al.* (1987).

out on other Late Palaeozoic formations in the same region, in order to test the hypotheses mentioned above, and thus to elucidate the tectonic evolution of this section of the Andean Chain during the Late Palaeozoic.

2 GEOLOGICAL SETTING

Figure 1 shows the main morphotectonic provinces of southwestern South America according to Hervé *et al.* (1987), and the location of the region where we have developed intensive palaeomagnetic studies on Late Palaeozoic rocks. Fig. 2 shows the geology of the Uspallata–Calingasta Valley region. This valley is the boundary between the morphotectonic provinces of the Cordillera Frontal and the Precordillera. Palaeomagnetic sampling localities also are shown on Fig. 2.

The oldest rocks exposed in this region are a highly deformed and mildly metamorphosed pre-Carboniferous (Devonian?) sequence of marine claystones and sandstones. A high angular uncomformity separates these rocks from the different sequences of Carboniferous glacial-marine and littoral sedimentary rocks that are broadly exposed in the Western Precordillera and Eastern Cordillera Frontal. These rocks have been assigned to many different formations, ranging in age from the Middle Carboniferous to the Lower Permian (Lopez Gamundi *et al.* 1987; Lopez Gamundi 1989; Gonzalez 1990). Only those formations from which palaeomagnetic data were obtained are shown in Fig. 2; all the others are refered to as undifferentiated Carboniferous sedimentary rocks. All these sequences constitute the fill of a north–south trending Middle Carboniferous–Lower Permian back-arc basin, called the 'Uspallata–Calingasta Basin' (Lopez Gamundi *et al.* 1987). The magmatic arc was situated westward during the Late Carboniferous, mainly in Chile (Hervé *et al.* 1987).

Figure 3 shows a simplified stratigraphic diagram of the Late Palaeozoic sequences exposed in the Uspallata–Calingasta Valley region. The stratigraphic positions of the sequences studied by palaeomagnetism within each formation also are shown. The studies carried out on the Majaditas (Loc. 1, Fig. 2), Yalguaraz and Portezuelo del Cenizo (Loc. 2, Fig. 2) Formations were reported previously (Valencio & Vilas 1985; Rapalini *et al.* 1989). Neither PP obtained from these three formations is coincident with the South American APWP.

During the Permian, magmatism migrated towards the east (Ramos *et al.* 1986) occupying also the old back-arc basins and producing very thick sequences of rhyolites, ignimbrites, tuffs and subordinate andesites. These sequences are known as the Choiyoi Group (Stipanicic 1965). In the Uspallata–Calingasta Valley region the Choiyoi Group is composed of a thin basal section (not thicker than 300 m) of flows, breccias and tuffs of andesitic to dacitic composition, represented by the Portezuelo del Cenizo Formation. This mesosilicic section is present at the base of the Choiyoi Group in many other areas of the Argentine central Andes (Cortés 1985). A radiometric date of 281 ± 10 Ma (Vilas & Valencio 1982; recomputed according to Steiger & Jagër 1977) indicates an early Early Permian age for this formation. An erosional uncomformity separates the lower mesosilicic section from the much thicker acidic sequences that make most of the Choiyoi Group. In this region the acidic terms are represented by the Tambillos and Horcajo Formations (Mirré 1966; Cortes 1985). The former comprises a 1700 m thick sequence of rhyolites, ignimbrites and pyroclastic breccias (Punta de Agua member, Fig. 3), together with 300 m of tuffs and lacustrine sandstones, claystones and limestones (Chiquero Member, Fig. 3) interfingered with the volcanic terms. These are conformably overlain by the Horcajo Formation to which belong the broadest extended exposures of the Choiyoi Group in this region (Fig. 2). This formation is a very thick sequence—more than 1500 m of ignimbrites, rhyolite flows and some related shallow intrusives. There are no radiometric data either for the Tambillos or for the Horcajo Formations in this area. Nevertheless, the stratigraphic relations (Cortes 1985) and some published radiometric dates from volcanic rocks of the Choiyoi Group exposed in other areas of the Cordillera Frontal and Precordillera (Caminos, Cordani & Linares 1979; Vilas & Valencio 1982) suggest a Late Permian–Early Triassic age for these rocks. As will be shown later the palaeomagnetic data obtained as well as a new Ar/K age for rocks presumably correlated with the Tambillos Formation suggest slightly older ages for these rocks (Fig. 3)

In the last decade the Choiyoi Group has been considered to be the volcanic product of a very extended magmatic arc

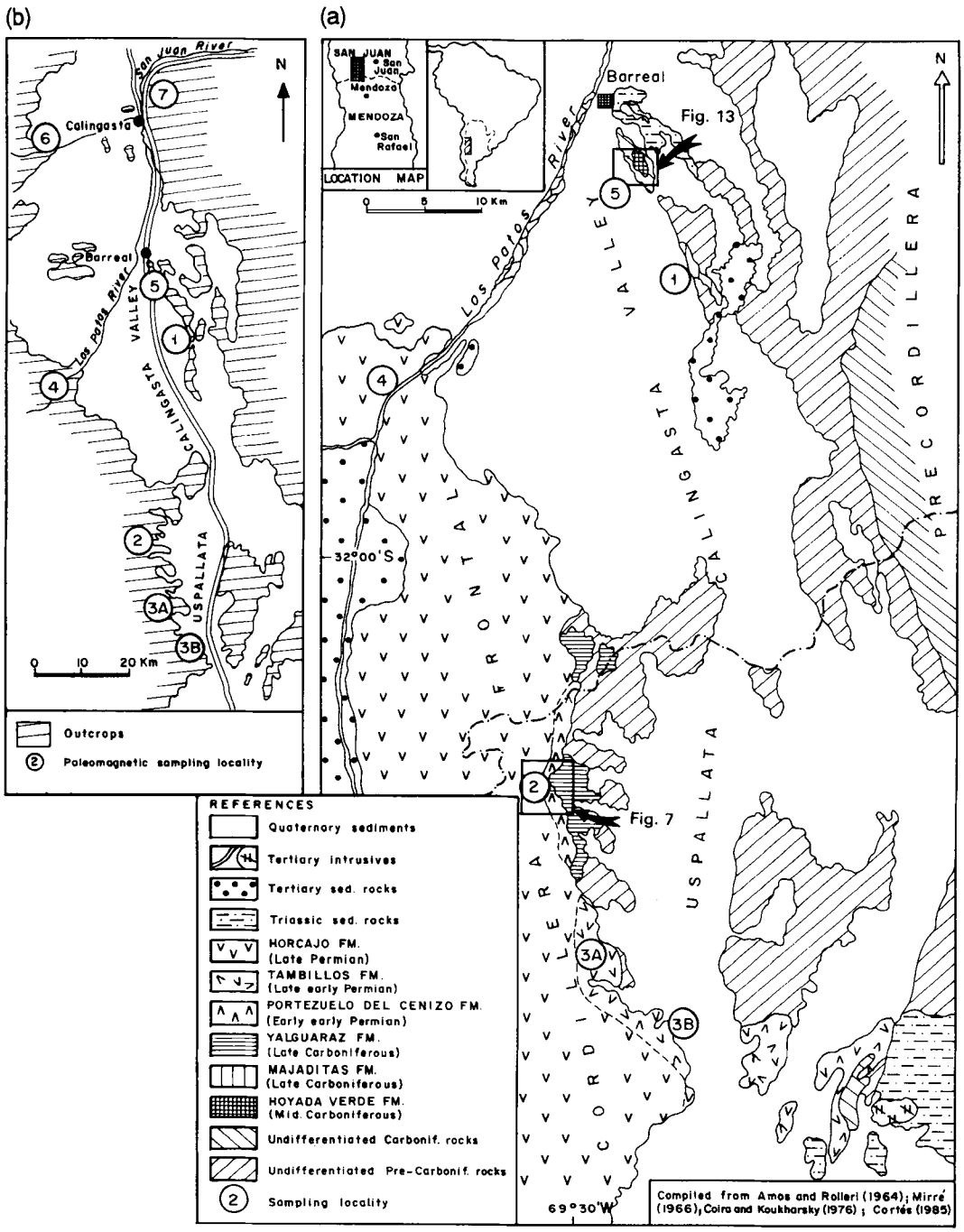


Figure 2. (a) Geologic map of the southern area of the Uspallata–Calingasta Valley, where the palaeomagnetic studies were carried out, and the location of the sampling localities. (b) Simplified sketch of the Uspallata–Calingasta Valley region, also showing the sampling localities 6 and 7 of two previous palaeomagnetic studies.

that occupied the western margin of southern South America during the Late Palaeozic (Frutos & Tobar 1975; Ramos *et al.* 1986; Hervé *et al.* 1987). Nevertheless recent geochemical data suggest that at least the upper acidic part of the Choiyoi Group sequence originated from extensional processes developed in a thickened continental crust (Kay *et al.* 1989). On the other hand, previous geochemical data from the Portezuelo del Cenizo Formation (Coira & Koukharsky 1976) show calk-alkaline affinities of the andesitic lower section of the Choiyoi Group. These rocks probably correspond to what Kay *et al.* (1989) called the Pre-Choiyoi rocks, with strong magmatic arc affinities.

3 PALAEOMAGNETIC STUDIES

3.1 The Tambillos Formation

The sampling localities of the Tambillos Formation are shown in Fig. 2 (3A and 3B). 27 hand samples from a 165 m thick sequence consisting of 14 units of ignimbrite,

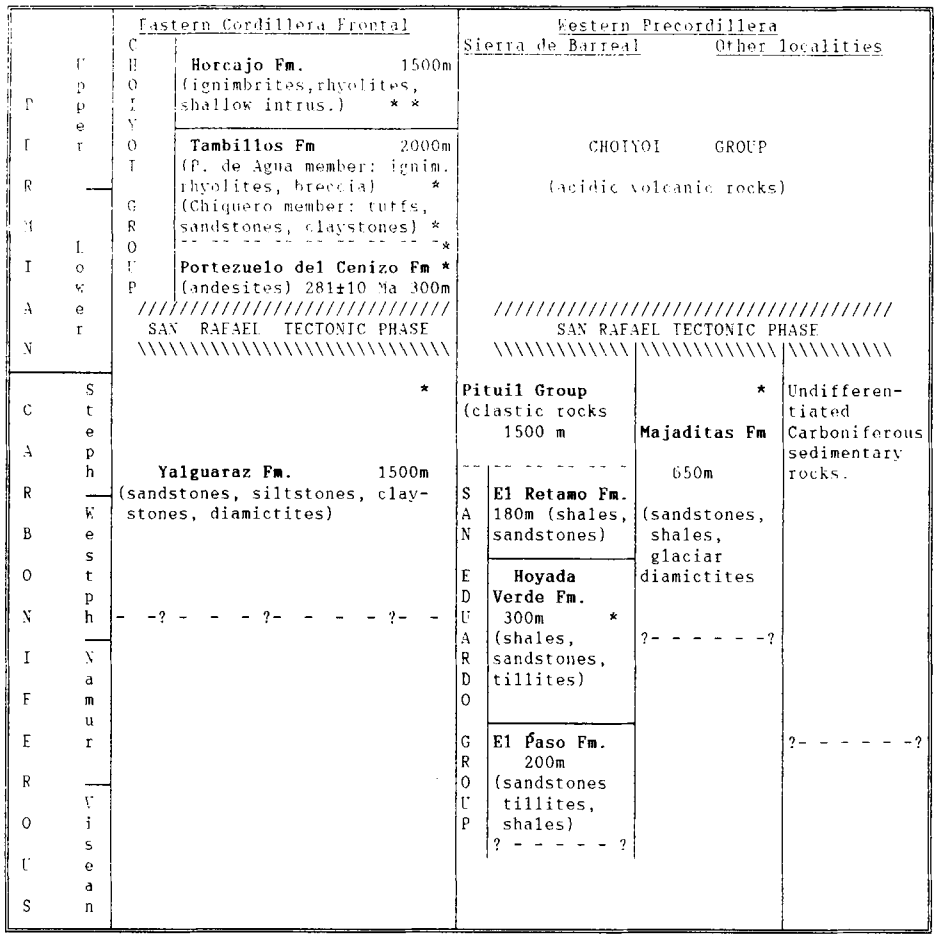


Figure 3. Schematic stratigraphic diagram of the Late Palaeozoic units exposed in the Uspallata–Calingasta Valley region. * indicates the number of localities and the stratigraphic position within each unit of the palaeomagnetic sampling. Simplified from Amos & Rolleri (1965), Lopez Gamundi *et al.* (1987) and Gonzalez (1990).

Table 1. Palaeomagnetic data of the Tambillos Fm. SRM: stable remanent magnetization. V.G.P.: virtual geomagnetic pole. * VGP considered of transitional polarity and excluded from the mean. Sample No means the identification numbers for the hand samples of each flow (i.e. 1/3 = 1, 2 and 3). Number of specimens used to compute the SRM for each flow is shown in brackets.

Sampling Locality	Flow No	Sample No	In situ SRM Dec.	Inc.	A95	Attitude Strike	Dip	Corr. SRM Dec.	Inc.	A95	Lat.S	V.G.P. Long.E	A95	
Locality 3A	1	1/3(6)	148.9	61.4		91	15	150.1	60.6		64.5	330.6		
	2	4-5(4)	160.3	64.1				161.2	63.2		70.8	334.8		
	3	6-7(3)	158.0	63.8				158.9	62.9		69.6	338.3		
	4	9(2)	197.1	50.5				196.9	49.5		75.4	23.6		
	5	10(2)	209.5	53.1				209.0	52.2		65.6	209.7		
	6	11-12(4)	167.9	62.9				168.6	62.0		75.8	326.8		
	7	13(2)	170.1	62.0				170.7	61.0		77.7	325.0		
	8	14-15(4)	168.4	71.9				169.4	71.0		65.7	305.3		
	9	16-17(4)	176.3	54.7				176.5	53.7		86.5	345.8		
	10	18-19(4)	161.8	54.1				162.4	53.2		75.2	10.1		
	11	20(2)	193.3	68.2				193.1	67.2		69.9	265.6		
	12	21-22(4)	177.5	59.3				177.8	58.3		83.1	304.8		
	13	23/25(6)	190.0	61.8				189.9	60.8		77.6	253.9		
	14	26-27(4)	167.4	69.7				168.3	68.8		68.4	310.3		
Mean Loc.3A	N=14		175.8	62.3	5			176.2	61.4	5	78.0	316.9	7	
Loc.3B	20	33-34(4)	195.3	32.7		348	19E	182.1	39.5		79.9	99.2		
	21	36(2)	124.5	45.3				114.6	30.9		29.3	18.6	*	
	24	41(1)	201.1	59.3				165.8	64.7		72.1	323.8		
Mean both localities	N=16		179.5	60.6	6			176.2	60.2	5	78.9	319.6	6	

ignimbritic breccia and rhyolite flows were collected from the Punta de Agua member (Fig. 3). Samples were taken along a roughly linear cross-section 1 km long, at locality 3A. The attitude of the sequence (Table 1) is almost horizontal.

The sequence sampled at locality 3B is 65 m thick and is composed of an upper section (16 m) of stratified tuffs and sandstones. Five hand samples from different stratigraphic levels were collected here. The lower 49 m of this sequence consists of five thick ignimbritic flows. Nine samples were collected from them. Both sections showed minor differences in their mean attitude (Table 1).

86 specimens, two and sometimes three from each sample, were subjected to demagnetization processes. One pilot specimen was chosen from each flow or stratigraphic level. All of these were submitted to stepwise AF demagnetization. At least 10 cleaning stages were applied in order to isolate the stable remanent magnetization (SRM). Another specimen from each flow was submitted to thermal demagnetization. Nine to 12 stages were usually applied until the specimens were completely demagnetized (Figs 4 and 5). For the remaining specimens the cleaning stages were determined on the basis of the magnetic behaviour of the pilot specimens.

Most samples from locality 3A (ET-1 to ET-27) carried a single positive inclination magnetic component with high coercivity (see specimen ET-4a, Fig. 4a). Some samples also carried an overprinted magnetic component with low coercivity and negative inclinations, which was interpreted as a probable viscous magnetization. Although some specimens became unstable beyond 40 to 50 mT, the AF demagnetization permitted the isolation of the SRM of most of the samples. The directions of the SRM were confirmed by submitting another specimen from each sample either to AF cleaning or in many cases to thermal demagnetization (specimen ET-4b in Fig. 4b). The latter was conclusive in those cases of magnetic instability beyond 40 to 50 mT. As Fig. 5 clearly shows the thermal demagnetization indicates that the remanence direction isolated at 40 mT, just before it became unstable, is the SRM.

The thermal cleaning gave some information on the ferromagnetic carriers. Fig. 4(b) shows three different blocking temperatures at 525 °C, 600 °C and 650 °C, probably corresponding to magnetite (or titanomagnetite),

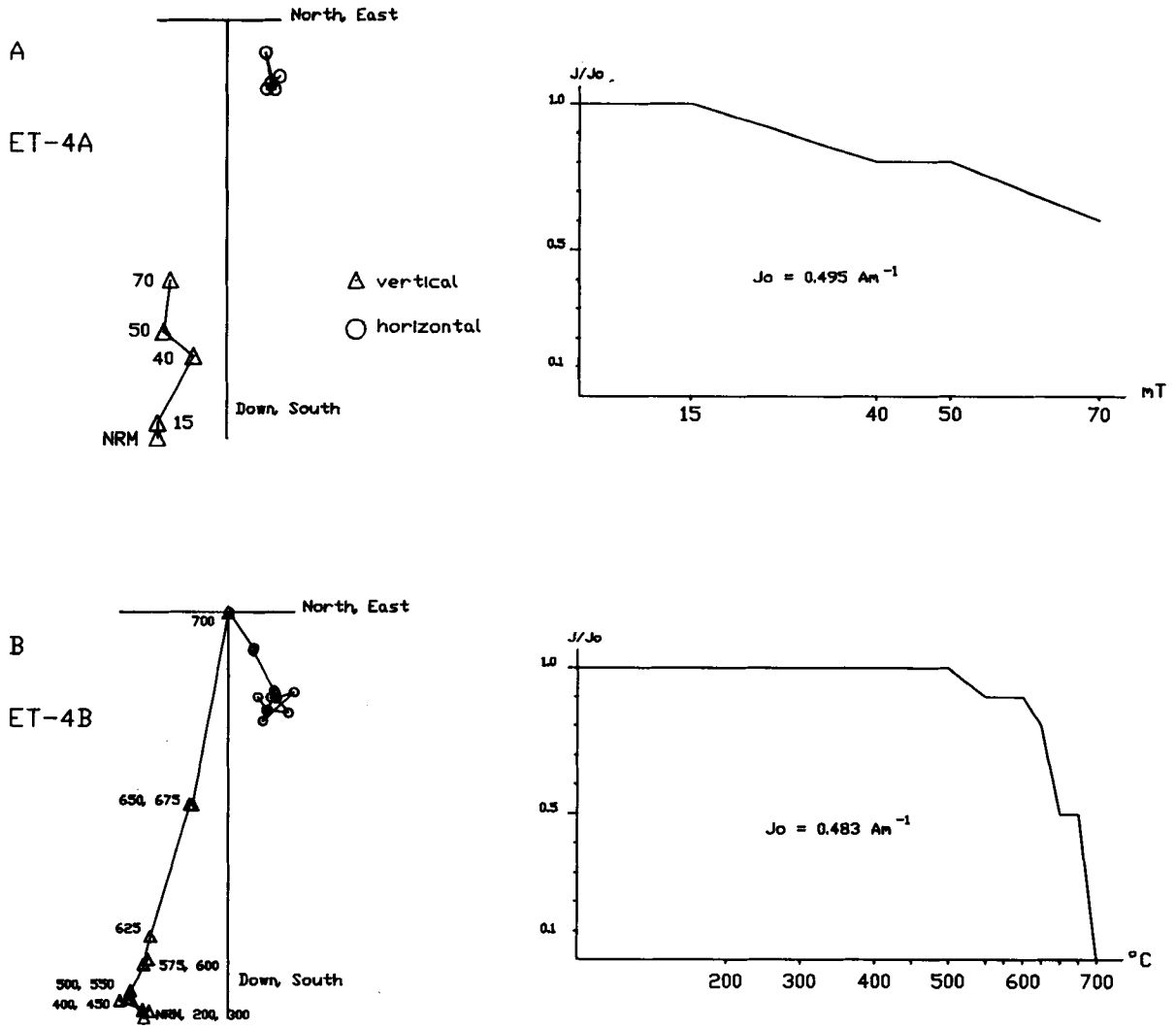


Figure 4. Typical magnetic behaviour of most of the samples of the Tambillos Formation, showing univectorial and very stable magnetization with high coercivity and positive inclination, during alternating field (A) and thermal (B) demagnetizations.

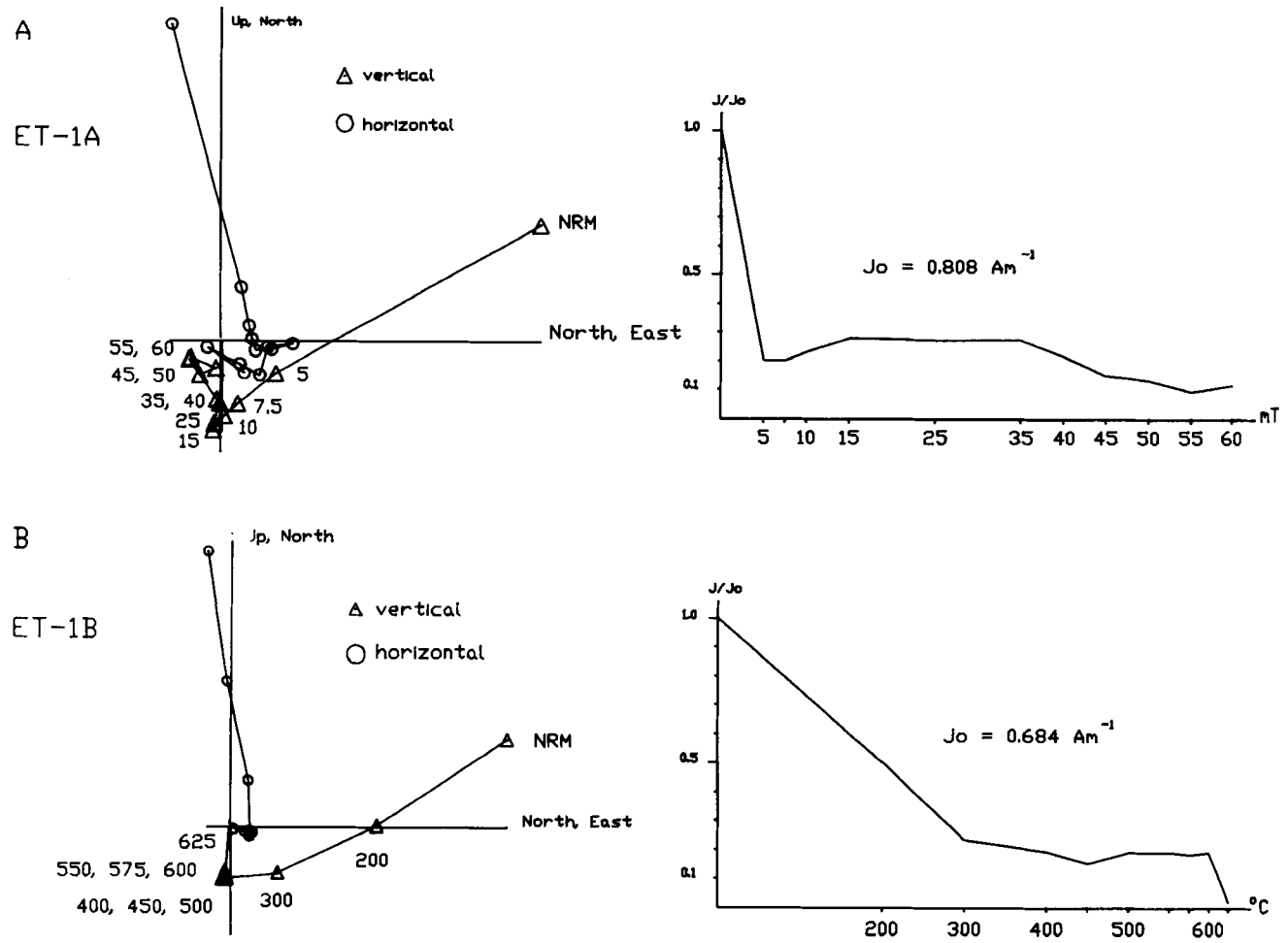


Figure 5. An example of those samples of the Tambillos Formation that showed an overprinted intense but soft negative magnetic component, after AF (A) and thermal (B) demagnetizations.

ilmeno-haematite (?) and haematite respectively. The orthogonal vector diagram shows that the three ferromagnetic components carry the same SRM direction. The coexistence of magnetite and haematite as carriers of the same stable remanence direction in these ignimbrites suggests that the haematite could be a product of primary oxidation at high temperatures during the volcanic process. On the other hand the demagnetization curve in Fig. 5 shows a fall near 300 °C, which is related to the elimination of the secondary magnetization, and another near 600 °C corresponding to the blocking temperature of the carrier of the SRM.

Many samples collected from the lower section of the sequence from site 3B were highly unstable under AF and thermal cleaning. This led us to rule out samples ET-35, 37, 38, 39 and 40, so that the SRM could not be isolated from the second and third ignimbritic flows of this sequence. The other samples (ET-33, 34, 36 and 41) showed magnetic behaviour similar to those mentioned for samples from locality 3A. No consistent characteristic stable magnetization could be isolated from samples from the pyroclastic and sedimentary beds exposed at 3B (upper section).

The SRM directions isolated for the specimens of each sample and for samples from each flow were averaged in order to obtain a mean SRM direction for each flow (Table 1). Fig. 6 shows these directions (a) *in situ* and (b) after bedding correction. Flows from locality 3A are represented by circles, while the three from 3B are shown by triangles. The small number of well-determined values of SRM isolated from 3B do not permit a fold test (Graham 1949; McFadden & Jones 1981). As a consequence, it is not possible to test whether or not the SRM is pre-tilting. However, the magnetic characteristics of most specimens, as

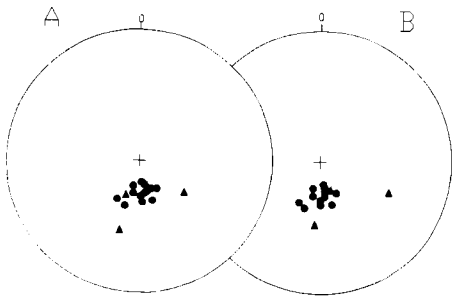


Figure 6. Stable remanent magnetizations (SRM) directions from each flow of the Tambillos Formation, before (A) and after (B) the structural corrections are applied. Solid symbols mean positive inclinations. Circles: SRM directions from sampling locality 3A; triangles: idem. from locality 3B.

well as the fact that the mean *in situ* SRM direction does not agree with the dipolar earth's magnetic field direction, suggest that the SRM isolated from these samples is primary. As most SRM directions belong to nearly horizontal beds (3A) the overall mean direction changes slightly after the bedding correction.

Virtual geomagnetic poles (VGP) for each flow are listed in Table 1. The number of flows sampled as well as some lithologic variation along the sequence suggest that enough time elapsed between the first and the last VGP to average the secular variation. The mean of all the computed VGPs is considered the palaeomagnetic pole for the Tambillos Formation (P15): 319.6°E, 78.9°S, $N = 16$, $k = 33.5$, $A_{95} = 6.5°$. In making this calculation all VGPs located over 40° from the mean were considered to be of transitional polarity and excluded from the average (McElhinny, Embleton & Wellman 1974).

3.2 The Horcajo Formation

This formation was sampled at two different localities (2 and 4, Fig. 2). A geologic map of locality 2 is shown in Fig. 7. D is the sampling area of the Horcajo Formation, while A, B and C are the sampling areas of the Yalguaraz and Portezuelo del Cenizo Formations (Valencio & Vilas 1985; Rapalini *et al.* 1989). An accurate correlation between the sections sampled at localities 2 and 4 cannot be made due to the great thickness and nearly homogeneous lithology of this formation as well as to the short lateral extension of individual ignimbritic flows. Both sequences originally had been assigned to different formations (Mirré 1966; Coira & Koukharsky 1976), although recently they have been integrated into the Horcajo Fm by Cortés (1985). Thus, the sequences are best assumed to be correlative only in a broad sense.

29 oriented hand samples from 16 ignimbrite, ignimbritic and pyroclastic breccia flows and welded-tuff beds were collected along a 210 m sequence at locality 2D (Fig. 7). The sequence shows little tectonic disturbance, its attitude is nearly homoclinal, with a shallow dip toward the north-northeast. 57 specimens (two from each hand sample, except sample AS-2) were submitted to progressive demagnetization processes.

28 hand samples from 15 ignimbrite and ignimbritic breccia flows and tuff beds were collected along a cross-section 2 km long, exposed at locality 4 on the western margin of Los Patos river (Fig. 2) This sequence, some 205 m thick, also is only slightly disturbed by tectonism, showing a homoclinal attitude with a shallow dip toward the southwest. 54 specimens from these samples were submitted to demagnetization.

The demagnetization procedures were very similar to that described for the Tambillos Formation. Both AF and thermal treatments were applied.

Samples from locality 2D showed two different magnetic behaviours. 70 per cent of the specimens carried a

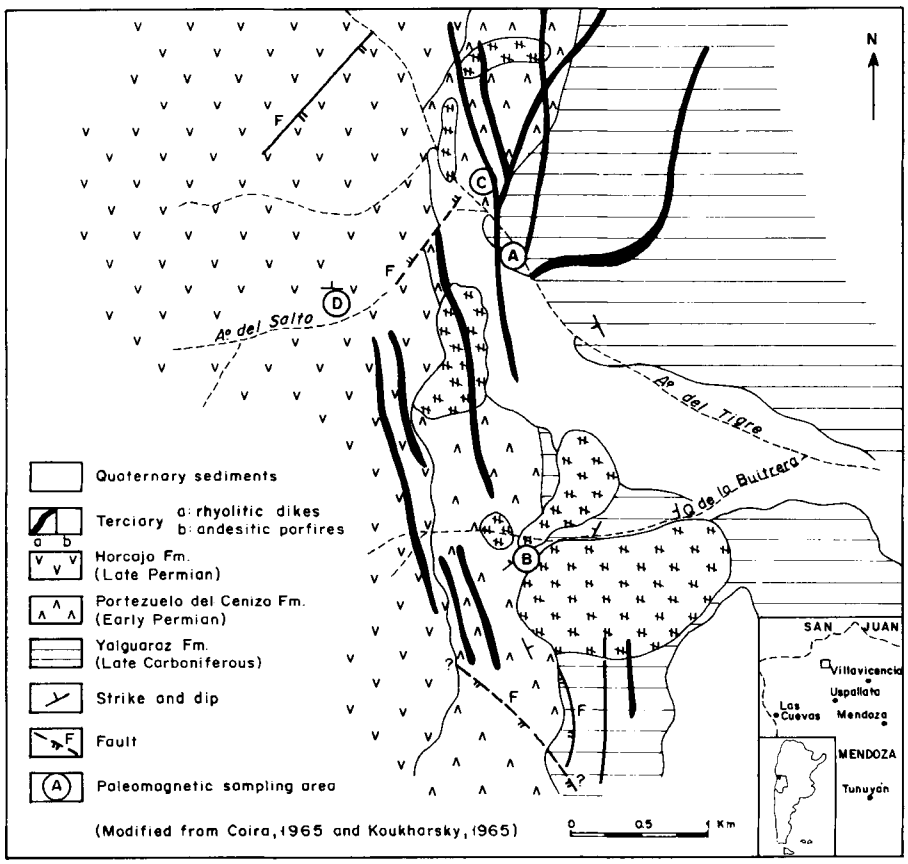


Figure 7. Geologic map of the Arroyo del Tigre area (sampling locality 2, Fig. 2) with the location of the sampling areas. (A) the Yalguaraz Formation (Valencio & Vilas 1985); (B and C) the Portezuelo del Cenizo Formation (Rapalini *et al.* 1989); (D) the Horcajo Formation (this paper).

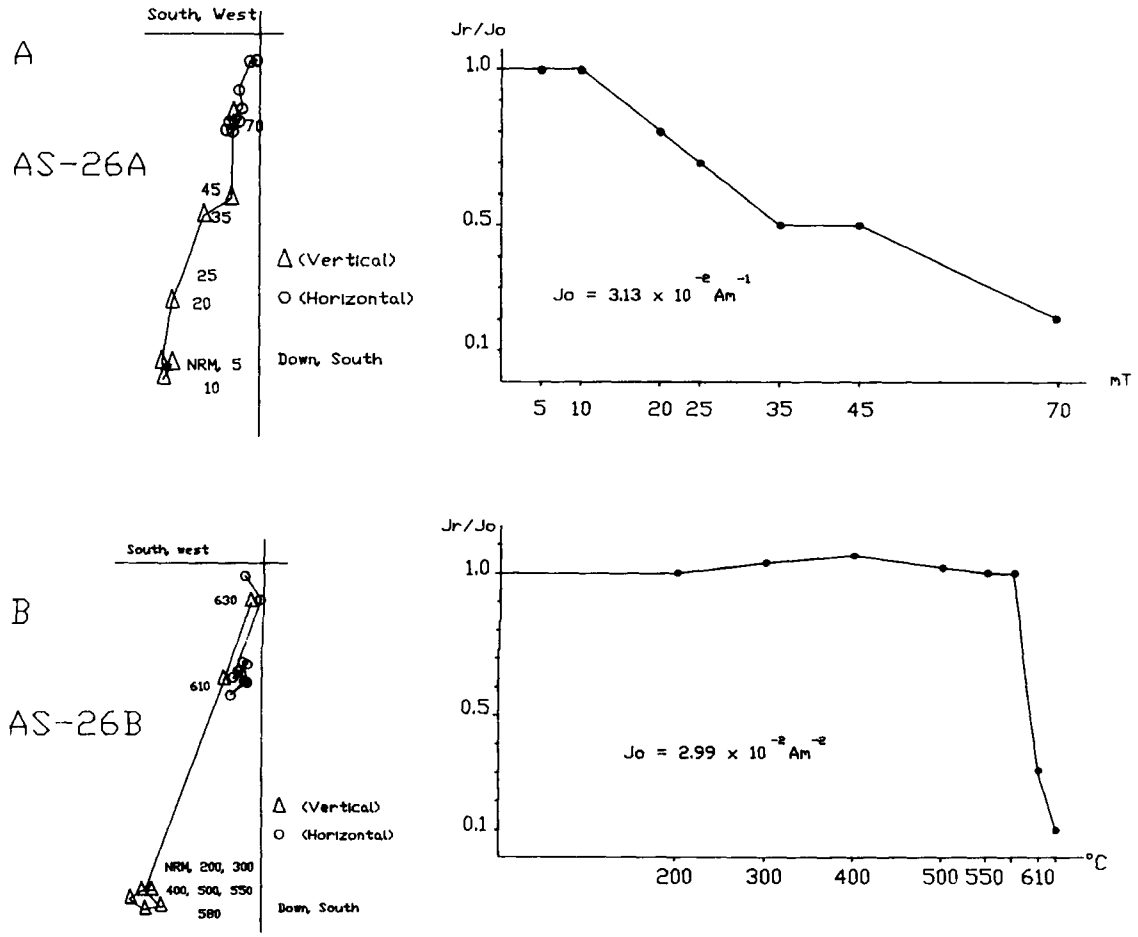


Figure 8. Representative magnetic behaviour of 70 per cent of the samples from locality 2D, showing univectorial SRM with high coercivities and positive inclinations, after (A) AF, and (B) thermal demagnetization. The latter suggests that haematite is probably the only magnetic carrier of the remanence.

univectorial SRM with high coercivity and positive inclination (Fig. 8). The high unblocking temperatures indicate that haematite (or perhaps low Ti ilmeno-haematite) is the most important magnetic carrier. The magnetic behaviour of the remaining 30 per cent is represented in Fig. 9. In these cases, the AF cleaning reveals only one magnetic component, with negative inclination and medium to low coercivity (Fig. 9a). By contrast, thermal demagnetization reveals the existence of another component with positive inclination, nearly antiparallel to the former. The low-temperature component is almost completely destroyed at 400 °C, while the other remains stable and is gradually destroyed up to the unblocking temperature(s) of the magnetic carrier(s). The negative low-temperature component thus is probably a secondary magnetization overprinting the stable positive remanence. The demagnetization curve of Fig. 9(b) also shows two unblocking temperatures at 580 °C and 650 °C, suggesting magnetite and haematite as the carriers of the remanence. Again, the existence of both minerals in these ignimbritic rocks each carrying the same remanence direction suggests that haematite could be a product of primary oxidation of magnetite at high temperatures.

Specimens from the Rio de los Patos area (locality 4) showed similar characteritics to those of the Arroyo del Tigre locality (2D). Most samples carried a unique stable magnetic component with positive inclination, and showed a response to the demagnetization treatment like that illustrated in Fig. 10 (LP-21a and 21b). The existence of two magnetic carriers with unblocking temperatures of 580 °C and 675 °C and moderate to high coercivities, suggest again magnetite and haematite. However, other samples had very high coercivities and a unique unblocking temperature around 675 °C (Fig. 11), suggesting haematite as the only magnetic carrier.

The SRM directions isolated for the specimens of each sample and for the samples of the same flow were averaged in order to obtain a mean SRM direction for each flow (or bed). In those cases in which the SRM showed no within-sample or within-flow consistency, the result was discarded.

Palaeomagnetic data for the Horcajo Fm are presented in Table 2. Fig. 12 shows the SRM directions from localities 2D (circles) and 4 (triangles), before (Fig. 12a) and after (Fig. 12b) bedding correction. The lack of coincidence of the mean directions from both localities either before or after the tilt correction could be due to the following three different reasons. (a) A substantially different age for both sequences, allowing the different directions to be due to apparent polar wander. This argument is weakened by

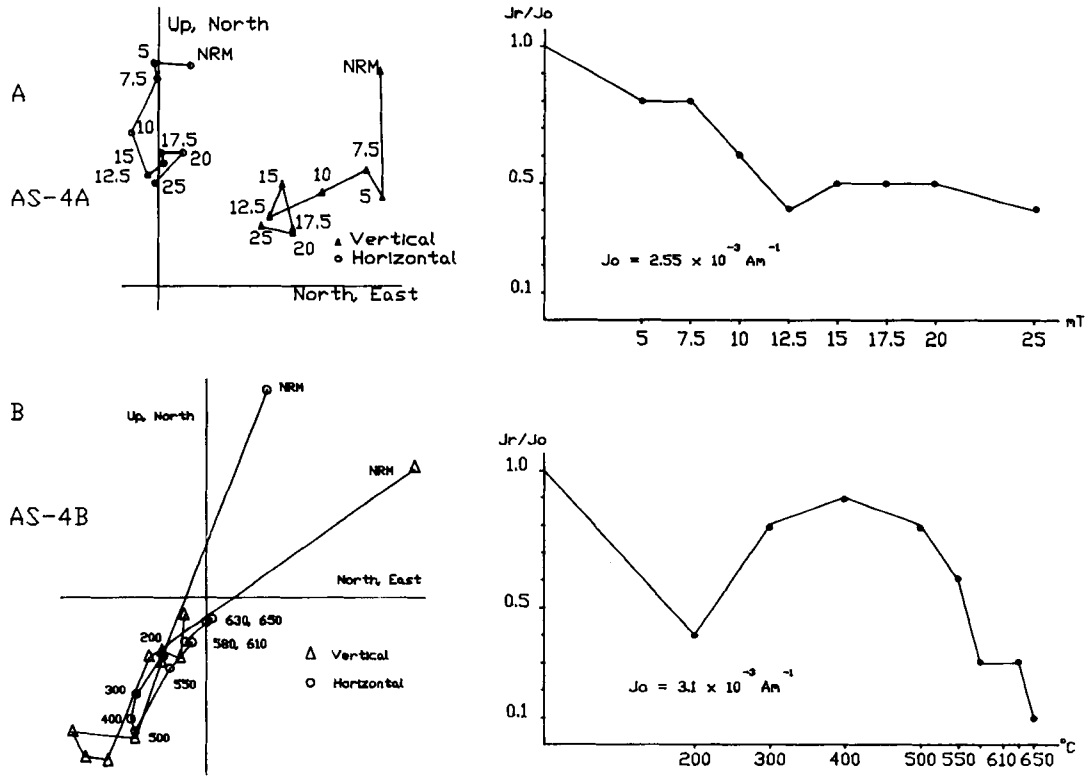


Figure 9. 30 per cent of the samples from locality 2D showed a superimposed negative magnetic component. (A) The AF cleaning failed to isolate the positive and more stable magnetization. (B) The thermal demagnetization easily removed the overprinted negative component at about 400 °C and indicated two unblocking temperatures, suggesting both magnetite and haematite as carriers of the remanence.

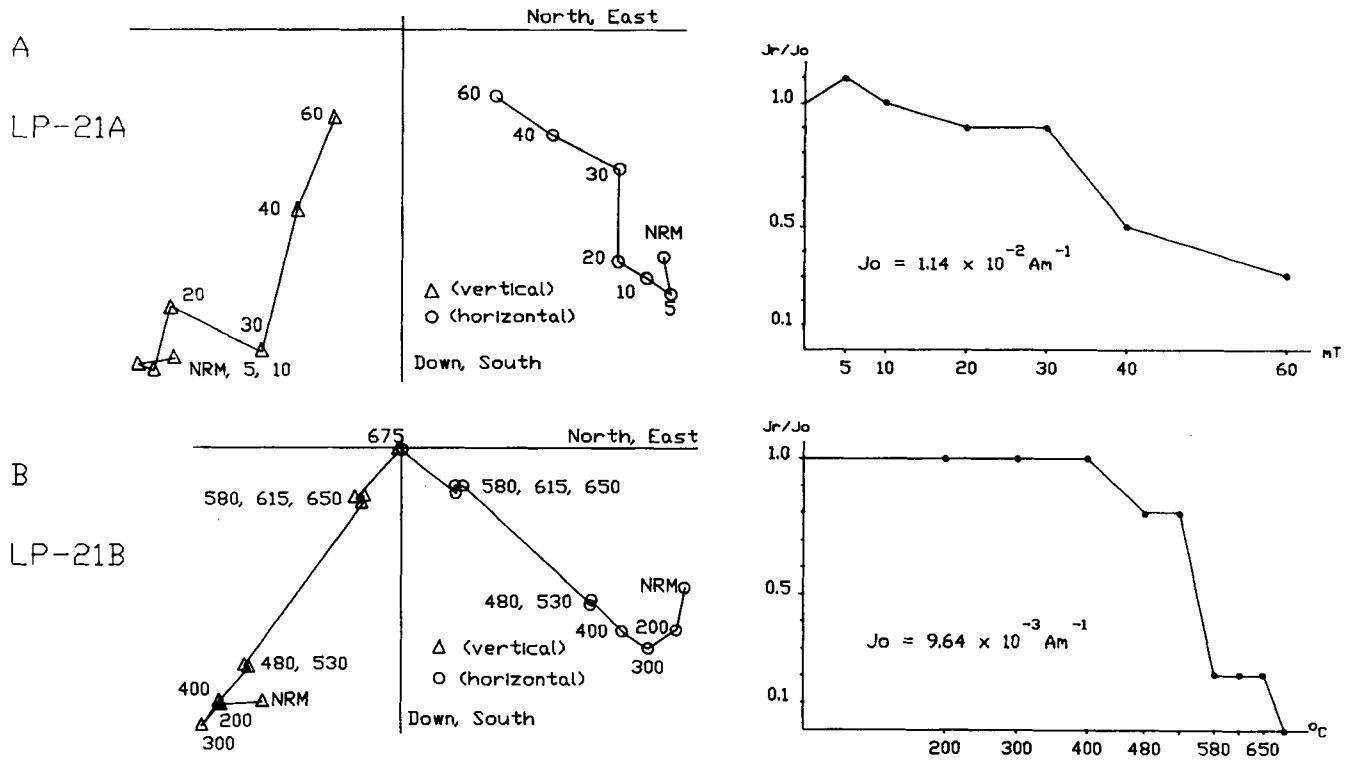


Figure 10. Representative magnetic behaviour of most of the samples from locality 4. The AF cleaning (A) shows a nearly univectorial magnetization with medium to high coercivities and positive inclination. The thermal demagnetization (B) confirms the univectorial decay of the magnetization and indicates at least two different magnetic carriers with unblocking temperatures of about 580 °C (magnetite?) and 675 °C (haematite?).

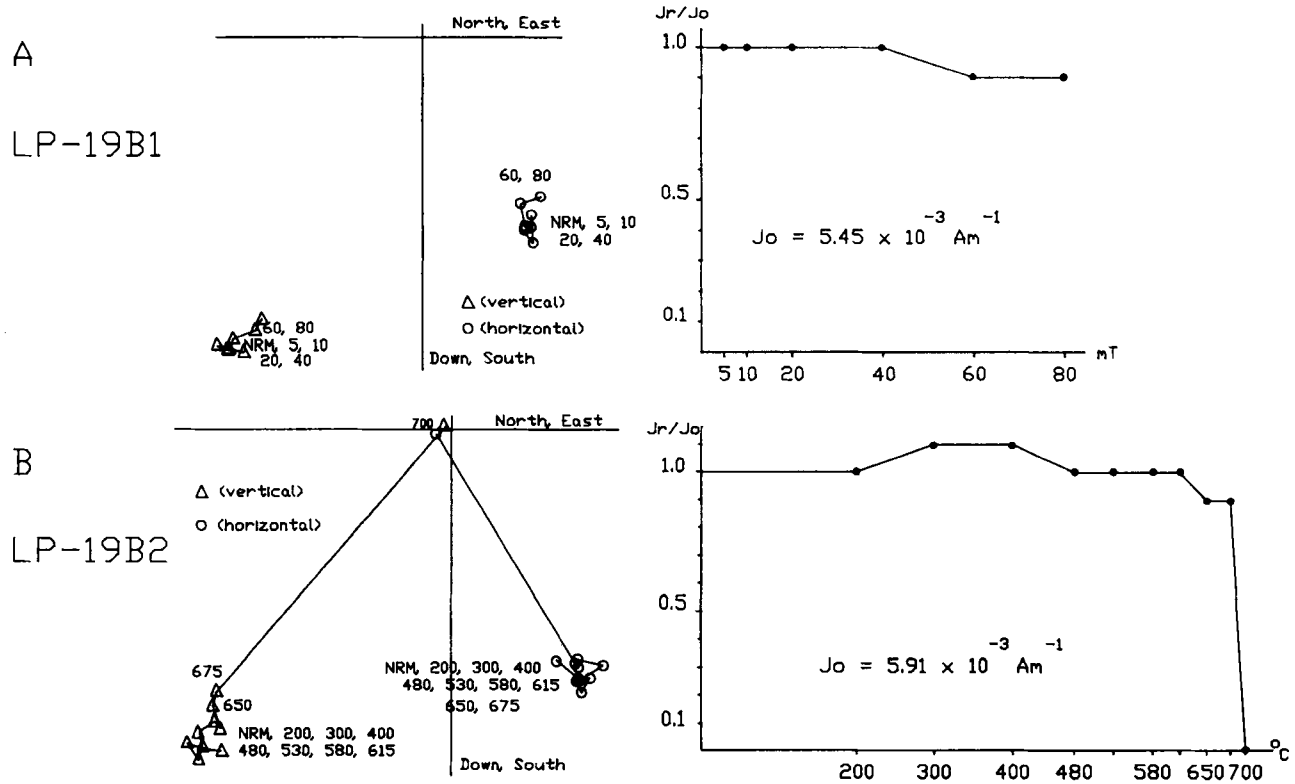


Figure 11. Representative magnetic behaviour of samples from locality 4 with extremely high coercivities (A). The thermal cleaning (B) shows only one magnetic carrier of the remanence, inferred as haematite due to its unblocking temperature being higher than 675 °C.

Table 2. Palaeomagnetic data from the Hocajo Fm. References as in Table 1.

Sampling Locality	Flow No	Sample No	In situ SRM Dec.	Inc.	A95	Attitude Strike	Dip	Corr. SRM Dec.	Inc.	A95	V.G.P. Lat.S	Long.E	A95	
Locality 2D	16	1/3(4)	212.4	67.9		249	11N	238.0	71.3		43.8	249.3		
	15	4/6(4)	193.4	52.3				203.7	59.6		69.3	230.7		
	14	7/9(6)	214.8	50.5				226.8	54.7		51.3	218.2		
	13	10(2)	152.5	78.3				138.6	88.2		34.8	293.3		
	12	11/13(5)	177.8	45.2				182.9	54.2		86.5	247.9		
	11	14-15(4)	236.3	62.8				255.9	62.6		31.2	235.8		
	10	16-17(4)	211.3	47.9				222.0	52.7		54.9	213.9		
	9	18(2)	195.4	51.0				205.4	58.0		68.4	225.1		
	8	19(1)	185.4	46.4				192.2	54.8		79.4	220.9		
	7	20-21(4)	207.1	70.6				237.0	74.7		43.6	256.7		
	6	22-23(4)	157.2	82.7				329.3	87.2		27.4	287.3		
	5	24(2)	146.4	66.9				140.3	76.7		49.4	315.3		
	4	25(2)	195.4	79.7				262.4	83.0		33.0	274.3		
	3	26-27(4)	192.4	72.9				225.3	79.1		45.4	269.1		
	2	28(2)	160.7	38.0				161.7	47.9		73.9	27.3		
	1	29(2)	186.4	65.0				203.1	72.8		59.7	266.3		
Mean Loc.2D		N=16	191.1	63.0	8			207.3	70.3	8	58.2	260.9	13	
Locality 4	15	18/20(6)	145.3	53.5		152	21W	172.7	50.9		83.8	20.5		
	14	21-22(4)	143.2	44.5				164.0	43.9		74.7	39.8		
	13	1-2,23-24(8)	150.4	41.0				167.9	38.3		75.1	60.9		
	12	3/5(6)	128.8	49.9				155.6	53.6		69.4	7.1		
	11	6(1)	151.7	34.4				165.5	32.0		70.5	64.7		
	10	7-8(3)	122.8	39.6				141.6	46.9		56.7	16.6		
	9	9(2)	162.3	30.6				173.1	24.8		70.2	90.0		
	7	12(2)		54.4-42.1					55.9-21.2		-34.6	11.4		*
	6	13(2)		112.4-44.1					101.2-29.0		1.1	39.3		*
	5	14-15(3)	190.9	19.9				194.6	6.1		58.1	138.8		
	2	27(2)	168.4	50.3				187.9	40.8		79.0	152.0		
	1	28(2)	356.6	80.0				271.0	71.0		25.2	251.4		
Mean Loc.4		N=10	152.2	47.5	16			173.5	43.4	16	83.2	56.7	18	
Mean Locs. 2D and 4		N=26	172.8	58.7	9			188.2	61.4	9	72.4	264.8	12	

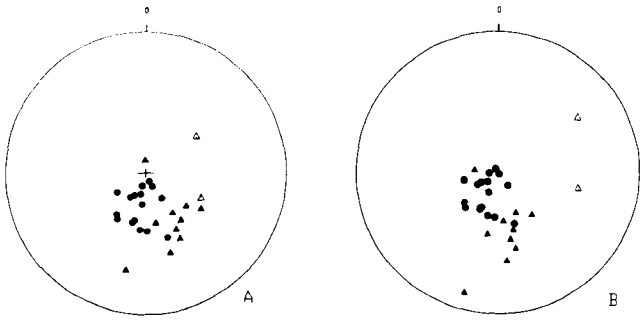


Figure 12. SRM directions for each flow of the Horcajo Formation, before (A) and after (B) the structural corrections are applied. Solid (open) symbols mean positive (negative) inclinations. Circles: SRM directions from locality 2D; triangles: idem. from locality 4.

geological evidence that suggests very close if not equal ages for both. (b) Incomplete averaging of secular variation in both sequences. This argument is weakened by the thickness of the sequences and the number of flows sampled. (c) Important errors in the definition of the palaeohorizontal due to the thick and irregular geometry of the ignimbritic flows, together with the problem of original dip of the flows. This is considered the most probable source of the disagreement of the mean directions from both localities. Wrong bedding corrections due to undetected tectonic movements of the sampling areas are not likely because most of the disagreement is in inclination (e.g., MacDonald 1980). Both arguments (b) and (c) suggest that the best procedure probably is to average the VGPs from both areas. This yields a single palaeomagnetic pole for the Horcajo Fm (P17): 264.8°E, 72.4°S, $N = 26$, $k = 6.4$, $A_{95} = 12°$.

3.3 The Hoyada Verde Formation

This formation (Mésigos 1953) is composed of 300 m of glacial-marine shales, claystones and sandstones. It has been assigned to the Namurian–Westphalian (Lopez Gamundi *et al.* 1987; Lopez Gamundi 1989; Gonzalez 1990) on the basis of its fossil assemblages.

Figure 13 is a simplified geologic map of locality 5, where a preliminary palaeomagnetic sampling of this formation was carried out on both flanks of the Hoyada Verde anticline (profiles B and C), following detailed sedimentologic logs previously performed by Lopez Gamundi (1983). 12 and six hand samples collected respectively along

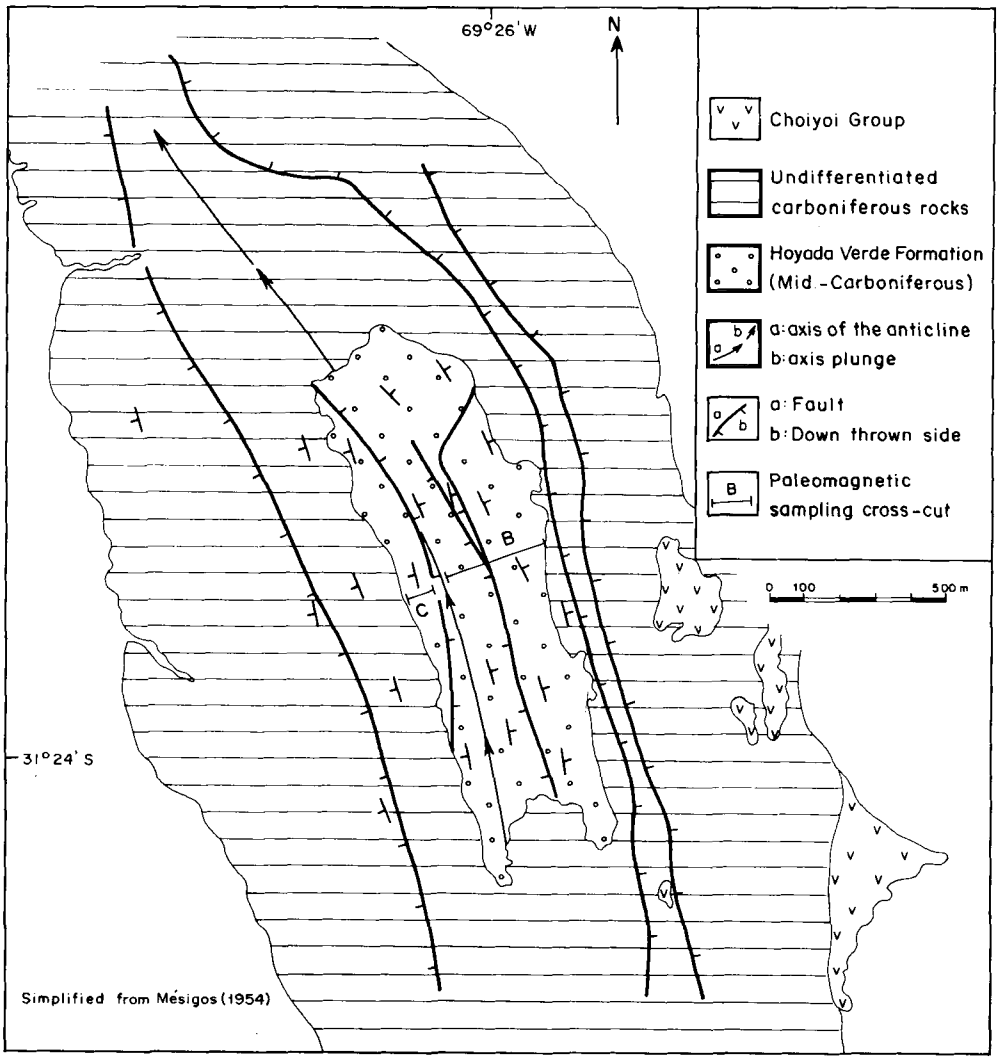


Figure 13. Geologic map of the Hoyada Verde anticline (sampling locality 5, Fig. 2) and the location of the sampling sections B and C.

them were processed. In agreement with their lithology, all showed very low natural remanent magnetization (NRM)—less than 1 mA m^{-1}. Two specimens from each hand sample were obtained and demagnetized. All were demagnetized by thermal cleaning at 100 °C, 200 °C. 300 °C, 400 °C, 450 °C and 500 °C. After this stage the intensity fell to the magnitude of the noise level of our flux-gate spinner magnetometer (UBA-8Hz, Vilas 1979). After each stage the bulk susceptibility of the specimens was measured in order to detect any chemical or mineralogical change due to heating.

Typical magnetic behaviour of these rocks is shown in Fig. 14. Specimen H4 is from one limb of the anticline (B, Fig. 13), while H45 belongs to the other (C, Fig. 13). Both specimens show similar response to thermal demagnetization, characterized by univectorial decay of the remanence after 200 °C. The increased directional dispersion at the last stages of demagnetization is due to the very low intensity of the remanence. Increase in intensity between NRM and 200 °C is caused by the destruction of an oppositely directed viscous magnetization. The low unblocking temperatures, near 450 °C, suggest that titanomagnetite is the most probable carrier of the magnetization.

Figure 15(a) shows the 95 per cent confidence circles of the mean SRM directions of each flank of the Hoyada Verde anticline, before (B and C) and after (B′ and C′) bedding correction; complete data are listed in Table 3. The mean directions from both limbs do not agree either before or after the correctons. However, Fig. 15(a) shows that partial correction certainly increased the agreement of the mean directions. Although these are preliminary results, this suggests a syntectonic magnetization of the Hoyada Verde rocks. Fig. 15(b) shows the value of the precision parameter k (Fisher 1953) for different percentages of partial structural correction (Miller & Kent 1986). For a k value greater than 37.08 the null hypothesis that both mean directions are the same cannot be rejected at 95 per cent confidence. It is obvious from the figure that this only happens for partial corrections of about 35 per cent. This confirms that the SRM of these rocks is secondary and was acquired during the folding of the Hoyada Verde anticline. The mean direction of the SRM, partially corrected (35 per cent), was then used to compute a palaeomagnetic pole for the Hoyada Verde Formation: C6, 356.2°E, 41.9°S, $d1 = 8.3°$, $d2 = 6.0°$; $d1$ and $d2$ are the semiaxes of the oval of 95 per cent confidence. This pole best represents the position of the palaeomagnetic pole during the folding of these rocks.

4 INTERPRETATION

Figure 16 shows a schematic Late Palaeozoic–Early Mesozoic apparent polar wander path (APWP) of South America and the positions of the recently obtained PPs: P15 (Tambillos Fm), P17 (Horcajo Fm) and C6 (Hoyada Verde Fm). Other Late Palaeozoic PPs obtained in this region: C7 (the Yalguaraz Fm, Late Carboniferous), C8 (the Majaditas Fm, Late Carboniferous), P 13 (the Portezuelo del Cenizo

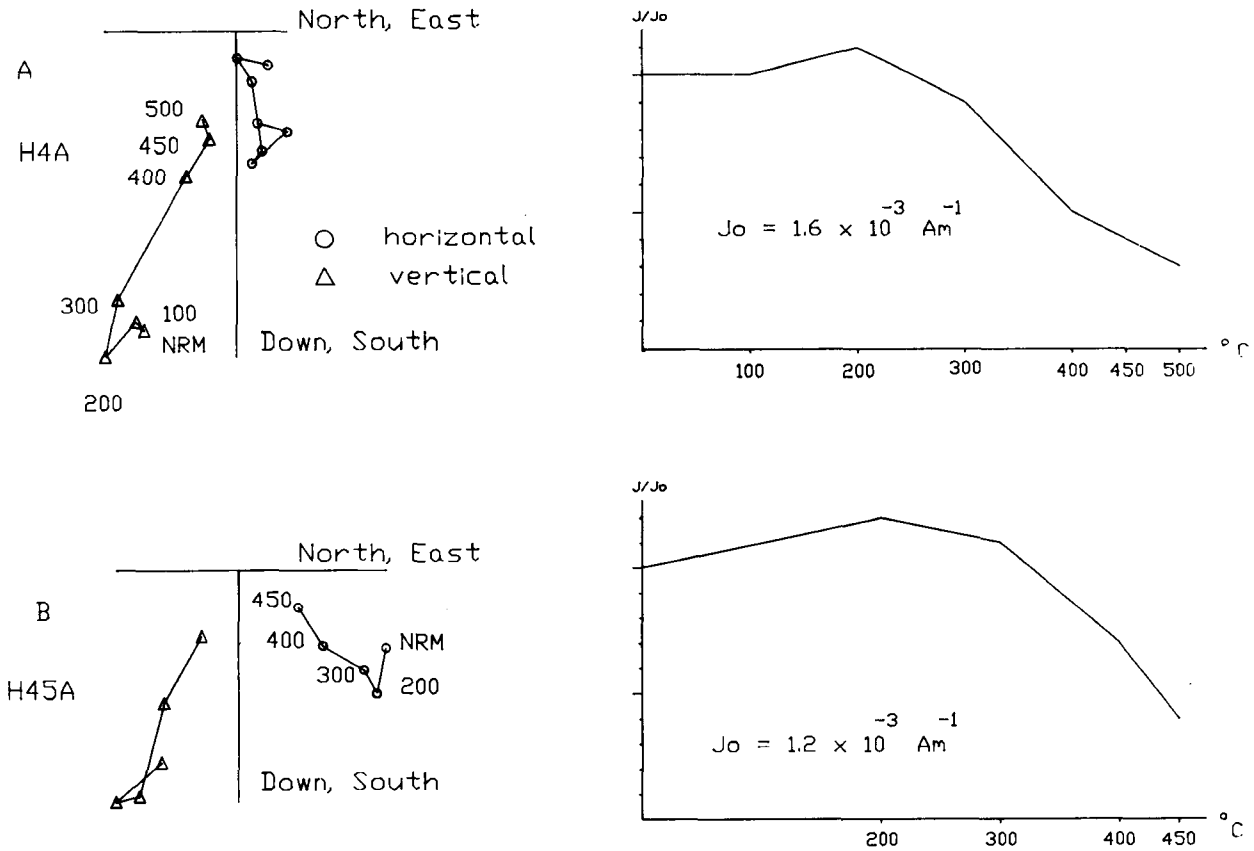


Figure 14. Representative magnetic behaviour of the samples from the Hoyada Verde Formation submitted to thermal demagnetization. (A) Representative sampled from the eastern flank of the Hoyada Verde anticline. (B) Idem. from the western flank. Note the nearly identical magnetic behaviour of both.

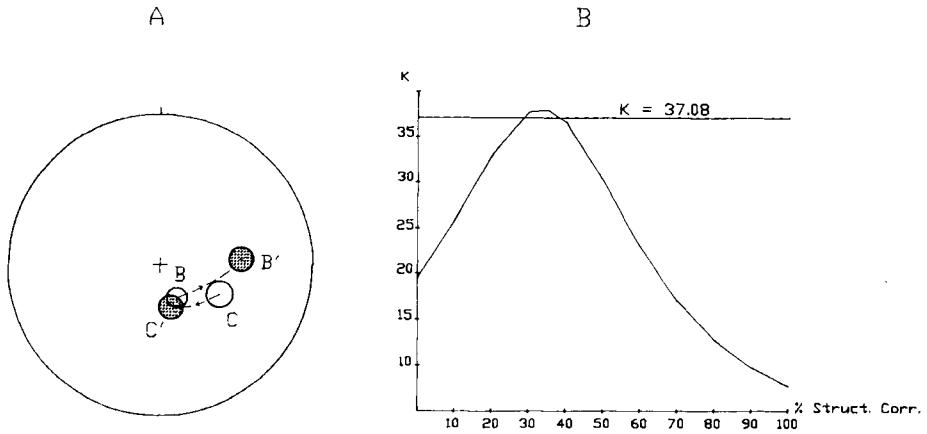


Figure 15. (A) The 95 per cent confidence circles of the mean SRM directions from each flank of the Hoyada Verde anticline before (B and C, open symbols) and after (B′ and C′, dashed symbols) the application of the structural corrections. The arrows show the small circles produced by these corrections. (B) Diagram of the statistical parameter K (Fisher 1953) of the SRM directions of both flanks versus the percentage of structural correction applied. It indicates that remanence of the Hoyada Verde Formation is probably synfolding. A value of K greater than 37.08 indicates that the null hypothesis that both mean directions are the same cannot be rejected with 95 per cent confidence.

Fm, 281 ± 10 Ma, early Early Permian) and P12 (the Choiyoi Group s.l., 267 ± 5 Ma, late Early Permian) as well as the mean PP of South America for the Late Carboniferous (CS) and the late Early Permian–Early Jurassic (PJ) according to Vilas (1981) are also shown. It is evident that C7, C8 and P13 are not coincident with the APWP of South America. As was mentioned above, two different hypotheses have been suggested to explain this: (i) *in situ* crustal block rotations or (ii) a displaced terrane.

P15, P17 and C6 are consistent with the APWP of stable South America. This is shown in Table 4 where values of rotation and poleward translation are computed for each PP of this region, following Beck (1989a).

The position of C6 is nearly coincident with the Late Carboniferous poles of South America. The small value of counterclockwise rotation when compared with CS is likely to be due to the simple unfolding method used, which assumes identical percentages of folding in both limbs. When different amounts of unfolding were applied to each limb the pole position became even closer to CS. In spite of that, the simplest hypothesis of equal unfolding was prefered as the actual mechanism of the folding is unknown. Nevertheless, the oval of 95 per cent confidence intersects some circles of confidence of other Late Carboniferous South American poles (Vilas 1981), not shown in the figure for clearness. This suggests that the folding of the Hoyada

Table 3. Preliminary palaeomagnetic data from the Hoyada Verde Fm. Palaeomagnetic pole computed from partially corrected (35 per cent) SRM. References as in Table 1.

Sampling locality	Sample No	In situ SRM Dec. (°)	Inc. (°)	N	k	A95 (°)	Attitude Strike (°)	Dip (°)	Corr. SRM Dec. (°)	Inc. (°)	N	k	A95 (°)	P.P. Lat.S (°)	Long.E (°)	D1 (°)	D2 (°)
Section B	H4	137.5	52.0				323	56	84.7	27.2							
	H5	168.0	65.5						82.8	40.8							
	H10	195.0	49.0						105.0	51.8							
	H13	133.0	54.5						91.0	21.8							
	H15	101.0	55.9						86.6	16.3							
	H17	114.0	68.0						73.6	21.6							
	H17R	181.5	64.5						82.4	46.6							
	H20	161.0	63.0						86.2	37.9							
	H27	133.0	67.0						78.7	27.3							
	H36	139.5	60.0						87.2	27.3							
	H37	171.0	66.0						81.9	42.0							
	H42	175.0	54.0						98.1	45.3							
Mean Values Section B		152.5	62.8	12	37.0	7.2	323	56	85.9	34.1	12	37.0	7.2				
Mean Values Section B (Corr.35%)									116.2	59.7	12	37.0	7.2				
Section C	H44	122.5	43.0				160	40	169.5	54.0							
	H45	122.0	43.0						169.2	54.3							
	H46	108.0	31.5						141.2	56.3							
	H47	97.0	46.0						157.9	71.6							
	H48	126.5	42.5						171.3	51.2							
	H49	123.0	49.0						179.2	56.3							
Mean Values Section C		116.4	43.0	6	62.8	8.5	160	40	165.6	57.8	6	62.8	8.5				
Mean Values Section C (Corr.35%)									128.5	51.6	6	62.8	8.5				
Mean Values B + C		136.4	57.5	18	19.5	8.0			102.7	47.2	18	7.6	13.4				
Mean Values B + C (Corr.35%)									120.9	57.1	18	38.0	5.7	41.9	356.2	8.3	6.0

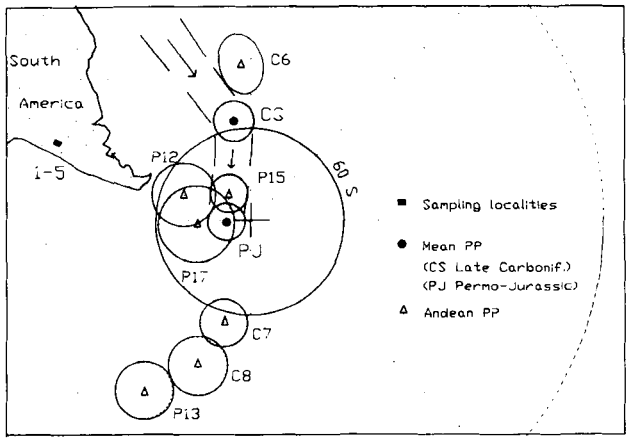


Figure 16. The Late Palaeozoic apparent polar wandering path of South America and the mean palaeomagnetic poles (PPs) for the Late Carboniferous (CS) and late Early Permian–Early Jurassic (PJ). The PPs from the Uspallata–Calingasta Valley region are shown by triangles with their respective 95 per cent confidence circle or oval. C7: Yalguaraz Fm and P12: Vega de los Machos Fm from Valencio & Vilas (1985). C8: Majaditas Fm and P13: Portezuelo del Cenizo Fm from Rapalini *et al.* (1989). C6: syntectonic magnetization of the Hoyada Verde Fm, P15: Tambillos Fm and P17: Horcajo Fm from this paper. See Vilas (1981) for complete references of the South American path.

Verde anticline occurred during the Late Carboniferous, and that little if any counterclockwise rotation of this area occurred since then. The concordant position of C6 also is strong evidence that this sector of the Argentine Andean Chain has been authochthonous since the Late Carboniferous. In previous papers (Valencio & Vilas 1985; Rapalini et al. 1989) the existence of a Late Palaeozoic displaced terrane was considered a possible interpretation of the anomalous PP positions, in spite of an admitted lack of geological evidence. Thus, of the two original hypotheses only *in situ* crustal block rotations remains. In this case C6 also indicates that a complex pattern of crustal block rotations took place in the Uspallata–Calingasta Valley region. Looking again at Fig. 2 it is evident that while localities 1 (C8) and 2 (C7 and P13) underwent very large clockwise rotations, locality 5 (C6) remained unrotated. Paleocurrent directions of the Hoyada Verde and Majaditas Fms (Gonzalez 1981; Lopez Gamundi 1983) are nearly north–south for the former and east–west to northeast–southwest for the latter. This disagreement can be resolved if the Majaditas block (Loc. 1) is rotated 80° (the minimum rotation computed for these rocks) counterclockwise. In this case the 'new' directions of the Majaditas paleocurrents become coincident with those from the Hoyada Verde Fm.

Table 4. Values of rotation (R), poleward translation (P) and the respective errors (ER and EP) computed for the Late Palaeozoic poles of the Uspallata–Calingasta Valley region. References for the poles in Fig. 16 and Tables 1 to 3.

Paleomagnetic pole	Reference pole	R	ER	P	EP
C6	CS	-20.7°	10.0	4.2°	7.8
C7	CS	75.3°	8.9	20.0°	7.4
C8	CS	98.7°	9.4	24.6°	8.0
P13	PJ	61.3°	7.3	15.4°	6.4
	P6	57.0°	7.5	0.9°	6.9
P12	PJ	8.6°	14.3	-16.3°	8.6
P15	PJ	-13.6°	8.0	-2.9°	8.1
P17	PJ	11.7°	13.9	-5.4°	13.9

P15 and P17 highly constrain the age of the rotations. One sampling area of the Horcajo Fm (D, Fig. 7) is located at locality 2. This was also the sampling locality of the Yalguaraz (A) and Portezuelo del Cenizo Fms (B and C), the PPs of which indicate large clockwise rotations. As there is physical continuity between the outcrops of the Horcajo Fm and the Portezuelo del Cenizo Fm, this could only be explained if the rotation occurred prior to the extrusion of the Horcajo ignimbrite flows.

The Horcajo Fm has been assigned to the Late Permian–Early Triassic (Cortés 1985) although no radiometric dates are available yet. The fact that all the VGPs computed from it are of reversed polarity (excepted two of possible transitional polarities, Table 2) suggests a Kiaman magnetic age, also consistent with the position of P17 close to other Permian poles of South America. Irving & Pullaihah (1976) placed the end of the Kiaman Superchron in the late Tatarian. This was recently confirmed by Molina Garza, Geissman & Van der Voo (1989) who suggested that it ended before 251 ± 4 Ma. This brackets the age of the Horcajo Fm as greater than 251 ± 4 Ma and less than 281 ± 10 Ma (the radiometric age of the Portezuelo del Cenizo Fm). This means a probable Late Permian age. Thus the crustal block rotations must have occurred between the early Early Permian (P13) and the Late Permian (P17).

The Tambillos Fm is older than the Horcajo Fm and its PP—P15—is also coincident with the Permian cratonic poles of South America (Table 4, Fig. 16). This could be explained by either rotation processes not affecting the Tambillos area or by rotation processes occurring prior to the extrusion of the Tambillos rocks. The latter is considered more probable because: (i) there is a very high lithologic and petrologic affinity between the Horcajo and Tambillos Fms rocks while the Portezuelo del Cenizo Fm rocks are markedly different; (ii) this lithologic and geochemical differences between the lower mesosilicic and the upper acidic terms of the Choiyoi Group may be related to an important change in the tectonic framework of the region (Kay *et al.* 1989; Mpodozis & Kay 1990); and (iii) there is a slight uncomformity under the Tambillos rocks, while the relationship between these and the Horcajo Fm is conformable (Cortés 1985).

If the rotations took place before the extrusion of the Tambillos ignimbrites, the age of rotation is even more tightly constrained. All of the VGPs of P15 show reversed polarity indicating also a probable Kiaman magnetic age. The Tambillos Fm must be older then than the Horcajo Fm which in turn is probably older than 251 ± 4 Ma; and must be younger than 281 ± 10 Ma. P12 is the palaeomagnetic pole of a sequence of tuffs and ignimbrites of the Choiyoi Group exposed in the Cordillera Frontal near the town of Calingasta, 50 km north from sampling localities 4 and 5 (Loc. 6, Fig. 2b). This sequence had been assigned to the Triassic (Valencio & Vilas 1985) due to a radiometric date of 233 ± 10 Ma (recomputed according to Steiger & Jagër 1977). However, a new K/Ar age of 267 ± 5 Ma (INGEIS, sample 2367) suggests loss of argon in the first sample. The sequence studied by Valencio & Vilas (1985) corresponds, according to a very detailed geologic survey accomplished recently (Caballé 1986), to the Vega de los Machos Fm. This, which also is part of the Choiyoi Group, presumably correlates with the Tambillos Fm, according to their similar lithologies and stratigraphic positions (Rapalini 1989a). In

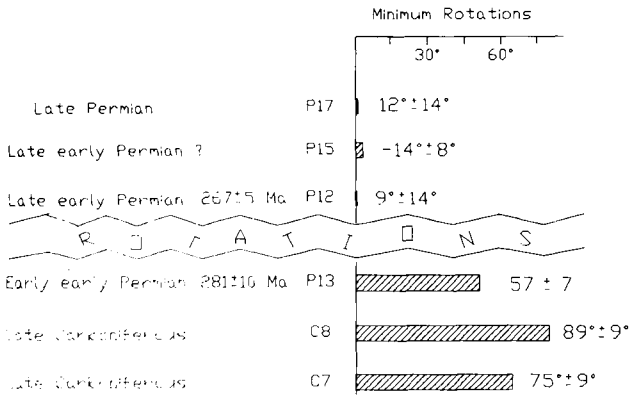


Figure 17. Diagram showing the age relationships of the rotations found in the Late Palaeozoic rocks of the Uspallata–Calingasta Valley region. According to this the main rotational event should have occurred in the time span between the early Early Permian and the late Early Permian.

this case a late Early Permian age for the Tambillos Fm is likely. This would even more constrain the age of the rotational process to between 280 and 265 Ma approximately. This is illustrated in Fig. 17.

4.1 Tectonic model

Mpodozis & Kay (1990) have recently suggested a correlation between the rotations reported here and the San Rafael tectonic event (Fig. 3), which they assigned to the collision of an unknown allochthonous terrane with the southwestern margin of Gondwana. However, most of the rotational process, if not all, seems to post-date this main tectonic event (compare Figs 3 and 17), being instead coincident with the development of a slight intra-Choiyoi erosional uncomformity (Cortés 1985; Caballe 1986). This seems to be the only field expression of the rotational event which then cannot be easily related to any compressional phase. The rotations must have been produced in a geologic framework where strike slip movements were the dominant ones. The Late Carboniferous–Early Permian magmatism of the Southern Andean Chain has been considered by most recent investigators to represent a magmatic arc on the southwestern continental margin of Gondwana (Frutos & Tobar 1975; Dalziel & Forsythe 1985; Ramos *et al.* 1986; Hervé *et al.* 1987; Mpodozis & Kay 1990; etc.). The Carboniferous sedimentary rocks are the fill of backarc basins located eastward of the volcanic arc (e.g. Claingasta–Uspallata Basin, Lopez Gamundi *et al.* 1987; Lopez Gamundi 1989). In this geotectonic setting probably most of the tectonic evolution was governed by the characteristics of the subduction process.

Fitch (1972) proposed a tectonic model to explain the generation of strike–slip faults parallel to active continental margins, by oblique subduction. Jarrard (1986) suggests that these faults are present in 50 per cent of present subduction zones, and two thirds if the overriding plate is continental. These strike–slip faults are located mainly in the areas of crustal weakness, such as the forearc or the volcanic arc [due to its higher geothermal gradient, Beck (1986)] and at distances ranging from 100 to 300 km from the oceanic trench. According to Jarrard the strike–slip zone is not necessarily a single fault but may be a diffuse zone characterized by minor faults, rotated crustal blocks and in echelon folds. Beck (1983, 1986, 1988a, 1989b) has investigated the parameters acting in crustal block rotations developed in active continental margins. The parameters are the angles of obliquity and dipping of subduction, the shape of the block, the efficiency of coupling and the relationship between the resistance to slip on the subduction zone to resistance to slip on the strike–slip fault parallel to the continental margin. These are also the parameters that define the detachment and transport of slivers along the margins. Beck (1988a) suggests that when the detachment is prevented or limited *in situ* block rotation is promoted. This would happen when the resistance to slip on the margin parallel faults is greater than the resistance in the subduction zone (more probable in new subduction zones) and with moderatly steep subduction angles. The block rotations will also be promoted when a 'buttress' prevents the displacement of crustal slivers along the continental margin. However, *in situ* rotations can take place simultaneously with detachment of margin slivers, but restricted to the detachment zone.

Figure 18 is a simplified sketch of the tectonic model proposed to explain the crustal block rotations found in the Uspallata–Calingasta Valley region. The sedimentary rocks of the Majaditas and Yalguaraz Fms (1) underlie the mesosilicic volcanic rocks of the Portezuelo del Cenizo Fm (2) which represents the volcanic arc. Several crustal blocks in this geotectonic setting undergo large clockwise rotations due to dextral strike slip movements parallel or subparallel to the continental margin. These, in turn, are caused by an oblique subduction of the Proto-Pacific Plate. The sketch does not exclude similar rotations nearer to the margin, in the forearc, including the detachment and displacement of slivers to the north. This should be tested by palaeomagnetism.

This tectonic model should have undergone a substantial change in the late Early Permian (Fig. 17) as the younger rocks are not rotated. This is coincident with the beginning of the late Early Permian–Middle Jurassic quasi-static period seen in South America APWP (Vilas 1981; Oviedo & Vilas 1984), and with the beginning of the extrusion of the thick rhyolithic sequences of the Choiyoi Group. Kay *et al.* (1989) recently have proposed that all these rocks belong not to a magmatic arc but to extensional processes in a thickened young crust and may be related to a nearly stationary state of Gondwana. According to Kay *et al.* (1989) this stationary period would be contemporary with an apparent lack of active subduction in this region. In our model the rotational processes obviously must have ended when the subduction ceased. Our model does not rule out the collisional origin of the San Rafael tectonic phase, as proposed by Mpodozis & Kay (1990). Moreover, the rearrangement of the subduction parameters after the collision of an allochthonous terrane may had led to the oblique subduction that caused tectonic rotations in the magmatic arc and backarc regions.

A relationship between these block rotations and a model of evolution of southern South American Late Paleozoic basins along NNW–SSE left-lateral strike–slip faults in an extensional regime (Nullo 1989, personal communication) does not seem likely. The sense of rotations (clockwise)

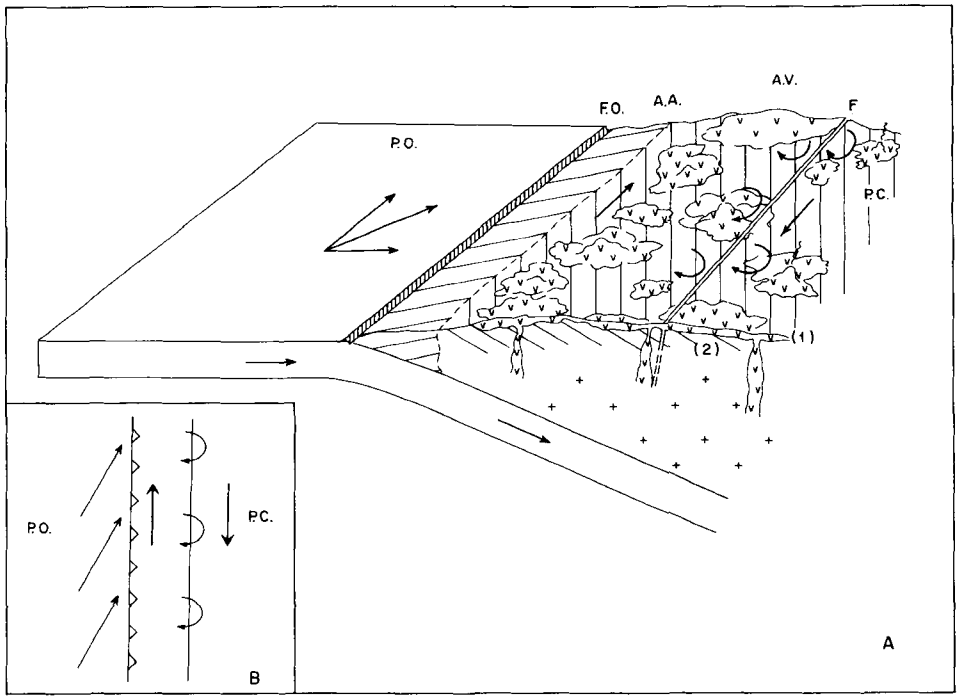


Figure 18. (A) Simplified diagram showing the tectonic model proposed to explain the rotations found in the andesitic rocks of the Portezuelo del Cenizo Fm (1) and the sedimentary rocks of the Yalguaraz and Majaditas Fms (2), in the Late Palaeozoic continental margin of southern South America. P.O.: Proto-Pacific oceanic plate; F.O.: oceanic trench, A.A.: fore-arc; A.V.: volcanic arc; F.: zone of strike-slip displacements (not necessarily a single main fault); P.C.: continental plate. Arrows show the sense of displacements and rotations. Complexities in the model arise from the co-existence of non-rotated crustal blocks among the rotated ones. (B) Simplified sketch of the development of crustal block rotations in continental margins with oblique subductions (modified from Fitch 1972).

suggests dextral strike-slip movements as a simpler mechanism, and the major extensional processes took place in southern South America in the Late Permian and Triassic (Uliana & Biddle 1987; Kay *et al.* 1989), while the rotations occurred in the late Early Permian. Moreover, several crustal block rotations in more recent times in the same continental margin (Garcia *et al.* 1988; Beck 1988b) were also probably caused by oblique subduction. The fact that the more recent rotations are restricted to a narrower zone along the continental margin (mainly the forearc) may be due to a different angle, obliquity and rate of subduction, as well as the existence of an older continental crust (Beck 1988a).

Northeastward oblique subduction during the Late Palaeozoic has also been proposed by Frutos (1988) who studied the stress-trajectories for the Chilean Andes at the same latitudes. For the Late Palaeozoic these trajectories are NNE–SSW which then led Frutos to suggest a NNE oblique subduction of the Proto-Pacific Plate (named by Frutos as 'Proto-Placa Farallón'). It is suggestive that these results lead independently to a similar model. A northward oblique subduction is also suggested by the development of an accretionary prism in southwest Patagonia during the Late Palaeozic—the shape of which is concave to the northeast (Forsythe 1982; Uliana & Biddle 1987)—and by the fact that the age relationships of these forearc complexes suggest a south to southwestern younging pattern of accretion and accreted material (Ling & Forsythe 1987).

The rotated crustal blocks probably are not restricted to this area of the Argentine Central Andes. Ordovician lavas of the Alcaparrosa Fm exposed 50 km north of this area (Loc. 7, Fig. 2b), east of the outcrops of the Vega de los Machos Fm (P12, Loc. 6), yielded a palaeomagnetic pole not consistent with coeval poles of stable South America and other Gondwanic continents (Vilas & Vlaencio 1978). Although several interpretations are possible, one of the simplest is that the rocks are situated in a crustal block that rotated 50° clockwise (Mena & Selles Martinez 1989). As the late Early Permian rocks of the Vega de los Machos Fm are unrotated the proposed rotation should have occurred prior to the late Early Permian, and are probably related to those found to the south.

Another clockwise rotated crustal block has been found by palaeomagnetic data on Permian rocks exposed several hundred kilometres south in the Neuquen province (39.1°S, 70.5°W, Rapalini *et al.* 1989). The rotational process has not been dated yet and it is considered that a different tectonic process during the Mesozoic led to this crustal block rotation (Rapalini 1989b). Nevertheless, a relation to Early Permian oblique subduction cannot be definitely ruled out.

5 PALAEOLATITUDE ANOMALIES

Table 4 shows significant palaeolatitude anomalies (poleward translation) for C7, C8 and P13 when they are compared with the corresponding mean South American PPs. As mentioned above, the possibility that these anomalous palaeolatitudes represent a displaced terrane has been ruled out both by geologic and palaeomagnetic data. A possible explanation of these results is that they represent an actual and very rapid palaeolatitudinal displacement of South America (probably Gondwana) during the Late

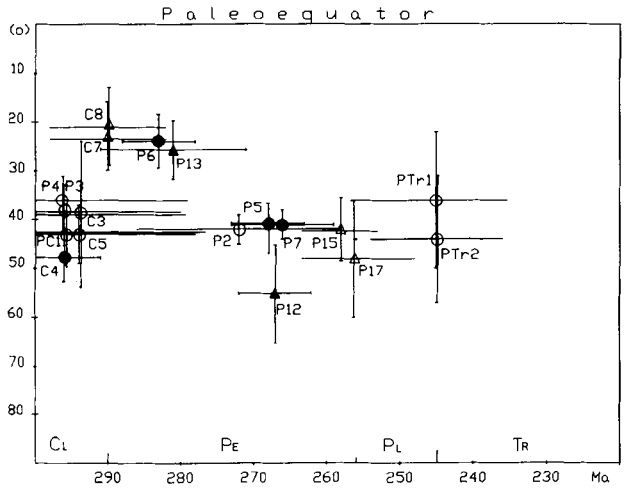


Figure 19. Diagram of latitude versus geologic time for the Uspallata–Calingasta Valley region according to the Late Palaeozoic South American poles. Triangles: poles from this region; circles: cratonic poles. Solid symbols indicate ages determined by radiometric datings. Horizontal and vertical bars indicate, respectively, the errors in age and paleolatitudes determinations, Cl: Late Caboniferous; Pe: Early Permian; Pl: Late Permian; Tr: Triassic. Complete references of PP in Vilas (1981), Rapalini *et al.* (1989) in this paper.

Carboniferous–Early Permian. This implies that the Late Carboniferous–Early Permian South American APWP is not accurately defined yet. Fig. 19 represents the palaeolatitudinal position of the Uspallata–Calingasta Valley region (at present at 32°S, 70°W) from Late Carboniferous to Late Permian, according to reliable South American palaeopoles. Those from stable areas are represented by circles while those from the Andean region are indicated by triangles. The age of each pole has been determined by radiometric (recomputed according to Steiger & Jagër 1977, when necessary), palaeontologic and/or magnetostratigraphic information. Those with radiometric dates are shown by solid symbols. The errors are shown by vertical and horizontal bars. The complete references for the cratonic poles can be found in Vilas (1981) and Oviedo & Vilas (1984). An age of 296 Ma has been assumed for C4 (296 ± 5 Ma) and other poles from the La Colina Fm in the Paganzo Basin (PC1, P3, P4), and a similar age for the sensu-lato Late Carboniferous poles of the Parana Basin (C3 and C5). The age of C7 and C8 has been assumed to correspond to the Carboniferous–Permian boundary according to the record of normal polarities in both the Yalguaraz and Majaditas Fms (Vilas & Valencio 1982; Rapalini *et al.* 1989) which could correspond to the Oak Creek normal event (Irving & Pullahiah 1976). Consistent with this, the only few fossils found in the Yalguaraz Fm indicate a Late Carboniferous–Early Permian age (Taboada 1985). P13 and P6 have nearly identical radiometric ages (281 ± 10 Ma and 283 ± 5 Ma) as well as the following P5, P7 and P12: 266 ± 5, 268 ± 5 and 267 ± 5 Ma, respectively. Fig. 17 shows significantly lower palaeolatitudes for the Carboniferous–Permian boundary and the early Early Permian (290–280 Ma.). It also shows a very rapid shift to subtropical latitudes (for the Uspallata–Calingasta Valley region) during the end of the Late Carboniferous and another rapid shift to moderate latitudes between approximately 280 and 270 Ma. As most of the poles that show the supposed latitudinal shift of South America are rotated no attempt can be made to suggest the actual trace of the APWP in the Early Permian. Nevertheless, the consistency with one pole from the stable areas (P6) supports this hypothesis. In this case the quasi-static period of South America would have begun in the late Early Permian after a rapid southward drift of the continent. Hypothetically this southward displacement of South America during the Early Permian would be consistent with a northward oblique subduction along its western margin as proposed above. This palaeolatitudinal displacement must be considered speculative until more early Permian data from stable South America and other Gondwanic continents are available.

6 CONCLUSIONS

Extensive palaeomagnetic studies carried out on Late Palaeozoic rocks in the Uspallata–Calingasta Valley region, in the Argentine Andes, have shown that during the Early Permian this region underwent large clockwise crustal block rotations. This process took place on the ancient southwestern margin of Gondwana and probably was related to northward oblique subduction of the Proto-Pacific plate. Before the Late Permian these processes ceased perhaps due to a major change in the characteristics of the subduction. This would almost certainly be related to the beginning of a stationary period of Gondwana and to an important variation in the magmatic processes developed in this region. The palaeomagnetic data obtained also have shown that the rotations did not follow a simple pattern and that unrotated blocks are present. Sparse palaeomagnetic data from other areas in the Argentine Andean Chain suggest that similar rotations could have occurred there, although they have not been dated yet.

ACKNOWLEDGMENTS

The authors wish to thank the Consejo Nacional de Investigaciones Cientificas y Técnicas and the University of Buenos Aires for their finnancial support. Dr O. Lopez Gamundi and Dr J. M. Cortés are acknowledged for their geological guidance as well as Lic. M. L. Bobbio who processed the Hoyada Verde samples. This is contribution no. 266 of the Laboratory of Palaeomagnetism 'Daniel A. Valencio'.

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